

Oleksiy Odarchuk

Backend Developer (Go) · Creator of the Piton language

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Performance-obsessed engineer with a product mindset. 1st place at BEST Hackathon 2026, 3rd at Mate Hackathon, invited to DOU Day 2026 as a Student with High Potential. Builds real products fast — from hackathon MVPs to production Go services.

🚀 WINS & HACKATHONS

🏆 1st Place — Lead Backend & Sensor Fusion

April 18–19, 2026

BEST Hackathon 2026 — GPS-less Drone Trajectory

- Algorithm to **reconstruct a UAV's trajectory without GPS** — from IMU, barometer and other inertial-sensor logs. Sensor fusion in **Go + C**: Go for the pipeline & integration, C inline for the hot numerical kernels.
- 1st place among top teams** — judges singled out the architectural resilience: modular split telemetry / integration / simulation let us swap datasets without rewriting.

🏆 3rd Place — Backend Engineer

March 14, 2026

Mate Hackathon (50 participants, 10 teams)

- Designed and built a Go backend from scratch in **7 hours** for an AI-powered MarTech tool that automates competitor-ad analysis.
- Dual AI pipeline: GPT-4o for marketing trigger analysis + Nano Banano 2 (fal.ai) for creative generation; backend routed all AI calls to cloud APIs, sustaining throughput under hackathon load.
- Beat teams with 5+ years of experience by focusing on a stable MVP — **3rd of 10 teams**.

Security Researcher → Job Offer

2026

College Blockchain System "Student Hryvnia"

- Discovered and exploited a critical vulnerability: frontend capped transaction amounts, **backend had no such check** — enabling arbitrary negative-value transfers. Reported responsibly; received a job offer from administration.
- Reverse-engineered the official JS miner through Webpack bundles — then built a faster Go replacement with a goroutine worker pool (SHMiner — p. 2).

📅 DOU Day 2026 — Student with High Potential

Personally invited by the organisers with a free ticket — a dedicated track for promising student developers.

🔧 TECH STACK

Languages: **Go** **Python** **C**

TypeScript

Backend: **Gin** **Fiber** **REST**

WebSockets

Goroutines

JWT

OAuth 2.0

Databases: **PostgreSQL** **Redis**

ClickHouse

Frontend: **Svelte** **Wails**

Tailwind

AI / APIs: **GPT-4o** **Fal.AI**

DevOps: **Docker** **Linux (Arch)**

Git

TinyGo

Bruno

Typst

🎓 EDUCATION & LANGUAGES

Network Technologies — IT

College NULP, 2nd year

2024 — present

IT Class — Science Lyceum, Rivne

2022 — 2024

Ukrainian — native · **English** — B1

Piton — My own programming language

[github](#) → [playground](#) → [docs](#) →

Ukrainian transliterated language + interpreter + flowchart generator

Interpreted language with an authentic Ukrainian syntax (in transliteration). Written from scratch in **Go without yacc/bison**: recursive descent parser, tree-walking evaluator, native SVG flowchart generation (D2), Turing-complete. Web playground: interpreter compiled to **WebAssembly via TinyGo** (220 KB vs. 24 MB with stock Go) + a separate Go-WASM module for D2 visualization (lazy-loaded).

[Go](#) [TinyGo](#) [WebAssembly](#) [AST](#) [D2 / SVG](#) [Lexer + Parser](#) [Nix Flakes](#)

OPEN SOURCE SDK

go-monobank-sdk — [github](#) →

Go SDK for every monobank API

Full coverage of **personal**, **provider** (with monoКЕП), **business** for legal entities, **acquiring** (subscriptions, T2P) and **Installment Purchases**. ECDSA webhook verification with auto-rotating keys, drop-in `http.Handler`, typed money/currency/MCC, HMAC-SHA256 for installment callbacks, a fake monobank server for tests, and a separate OpenTelemetry sub-module.

[Go](#) [ECDSA](#)

[HMAC-SHA256](#)

[OpenTelemetry](#) [iter.Seq2](#)

SDK CONSUMER

monokasa — Telegram [github](#) →

Ticket-Selling Bot

Sells tickets to a single show through a monobank jar and issues PDFs with QR codes scanned at the venue entrance. Live consumer of my own **go-monobank-sdk**: webhook auto-registers at startup, reservations get a 15-minute hold, and show-time reminders fire at-most-once even across restarts.

[Go](#) [SQLite](#) [telebot](#)

[PDF + QR](#)

[go-monobank-sdk](#)

1000× FASTER

SHMiner — Desktop [github](#) →

Crypto Miner

Cross-platform desktop miner built with **Wails (Go + Svelte/TypeScript)**. Rewrote SHA-256 hashing from JS to Go with a goroutine worker pool — achieved **1000× faster hashrate** than the official client on identical hardware. AES-encrypted wallet storage, real-time WebSocket dashboard, multi-wallet, zen mode.

[Go](#) [Svelte](#) [TypeScript](#)

[Wails](#) [WebSockets](#)

[Goroutines](#)

150+ USERS

AbitAssistant Bot [github](#) →

Telegram bot helping Ukrainian students track university admission rankings. Reverse-engineered closed third-party APIs to fetch live competition data. **150+ active users in first year**, marketing campaign planned for 2026.

[Python](#) [Aiogram](#)

[PostgreSQL](#) [BeautifulSoup](#)

[Pandas](#) [Docker](#)

LinkShortener Bot [github](#) →

Telegram URL shortener with QR generation and geo analytics. PostgreSQL for storage, Redis for redirect caching, ClickHouse for real-time click analytics (country, city, platform). Docker, GeolP2.

[Go](#) [PostgreSQL](#) [Redis](#)

[ClickHouse](#) [Docker](#)

[Telegram API](#)

ToDoList Pro — Go + [github](#) →

Svelte

Production-grade task manager: Go/ Gin backend with clean architecture (repository → service → handler), JWT + OAuth 2.0 (Google), email verification, Swagger docs. Svelte 5 + TypeScript + Tailwind. Multi-stage Docker build.

[Go](#) [Gin](#) [Svelte](#)

[TypeScript](#) [JWT](#)

[OAuth 2.0](#) [PostgreSQL](#)

[Docker](#)